Phys 515 HW 1 - Andrew Koren

1. show $\frac{\kappa_S}{\kappa_T} = \frac{C_V}{C_p}$

$$\kappa_S = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_S, \qquad \kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T,$$
$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \qquad C_p = T\left(\frac{\partial S}{\partial T}\right)_p.$$

so

$$\frac{\kappa_S}{\kappa_T} = \frac{\left(\frac{\partial V}{\partial p}\right)_S}{\left(\frac{\partial V}{\partial p}\right)_T}.$$

We need some $\partial S/\partial T$'s to to get C_V and C_p, so we'll use the triple product rule to excrete ∂S and ∂T:

$$\left(\frac{\partial V}{\partial p}\right)_S = -\frac{\left(\frac{\partial S}{\partial p}\right)_V}{\left(\frac{\partial S}{\partial V}\right)_p}$$
$$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{\left(\frac{\partial T}{\partial p}\right)_V}{\left(\frac{\partial T}{\partial V}\right)_p}$$
$$\frac{\kappa_S}{\kappa_T} = \frac{\left(\frac{\partial S}{\partial p}\right)_V}{\left(\frac{\partial T}{\partial p}\right)_V}\frac{\left(\frac{\partial T}{\partial V}\right)_p}{\left(\frac{\partial S}{\partial V}\right)_p} = \left(\frac{\partial S}{\partial T}\right)_V \frac{1}{\left(\frac{\partial S}{\partial T}\right)_p} = \frac{TC_V}{TC_p} = \frac{C_V}{C_p}$$

2.

Polytropic -> constant C.

We want to derive an equation s.t.

$$pnV^{n-1}dV + V^n dp = 0$$
$$\frac{dp}{p} + n\frac{dV}{V} = 0$$

and $n = \frac{C-C_p}{C-C_V}$ has to get in somehow.

Heat capacity $dQ = dU + pdV = CdT$, Ideal gas $dU = C_V dT$, so

$$pdV = dQ - dU = (C - C_V)dT$$

For C_V, use ideal gas definition

$$pV = (C_p - C_V)\,T$$
$$pdV + dpV = (C_p - C_V)\,dT$$

Now jam the expressions together

$$\frac{pdV + dpV}{(C_p - C_V)} = dT$$
$$pdV = \frac{(C - C_V)}{(C_p - C_V)}\,(pdV + dpV)$$
$$\frac{dV}{V} = \frac{(C - C_V)}{(C_p - C_V)}\left(\frac{dV}{V} + \frac{dp}{p}\right)$$

umm

$$(C_p - C_V)\,\frac{dV}{V} = (C - C_V)\frac{dV}{V} + (C - C_V)\frac{dp}{p}$$
$$0 = (C - C_p)\frac{dV}{V} + (C - C_V)\frac{dp}{p}$$
$$0 = \frac{(C - C_p)}{(C - C_V)}\frac{dV}{V} + \frac{dp}{p}$$

got it

- Adiabatic $dQ = 0 \to C = 0$ so $n = \frac{C_p}{C_V}$
- Isothermal $dT = 0$ so $pV = RT$ (const) and $n = 1$
- Isobaric $dp = 0$ so $ndV = 0$. Volume changes so $n = 0$
- Isochoric $dV = 0$ so $0 = \frac{C - C_V}{C_p - C_V}\frac{dp}{p}$. p changes so $C = C_V$ and $n = 1/0$

3. Reversible process -> Clausius equality. Isotherm can be reversible? Not sure about polytrope.

$$\oint \frac{dQ}{T} = 0$$

Integrate around ACBD

$$\int_{ACB} \frac{dQ}{T} + \int_{BDA} \frac{dQ}{T} = 0$$
$$\int_{\text{isotherm}} \frac{dQ}{T} + \int_{\text{polytrope}} \frac{CdT}{T} = 0$$
$$\int_{\text{isotherm}} \frac{dQ}{T} = -C \ln\left(\frac{T_A}{T_B}\right)$$

BUT if T_A and T_B are connected by an isothermal curve, $T_A = T_B$ so we have

$$\int_{\text{isotherm}} \frac{dQ}{T} = 0 = \frac{Q}{T}$$

However for an isotherm we have $Q = nRT \ln\left(\frac{V_A}{V_B}\right) = nRT \ln\left(\frac{P_B}{P_A}\right)$. Since $Q = 0$ also force $V_A = V_B$ and $P_A = P_B$ the two crossing points are both the same point and this is a contradiction. The isotherm and polytrope can only cross at one point.

4. a) Stefan Boltzman $u(T) = \sigma T^4 \rightarrow \frac{du}{u} = 4\frac{dT}{T}$

we have:

$$U = Vu(T), \quad p = \frac{1}{3}u(T)$$

we need: du, T, dT

$$\begin{aligned} dU &= dVu + Vdu \\ TdS - pdV &= dVu + Vdu \\ TdS &= \frac{4}{3}dVu + Vdu \\ dS &= \frac{4}{3}\frac{u}{T}dV + \frac{Vu'}{T}dT \end{aligned}$$

We can make use of dS through

$$\begin{aligned} dS &= \left(\frac{\partial S}{\partial V}\right)_T dV + \left(\frac{\partial S}{\partial T}\right)_V dT \\ \frac{\partial}{\partial T}\left(\frac{\partial S}{\partial V}\right)_T &= \frac{\partial}{\partial V}\left(\frac{\partial S}{\partial T}\right)_V \\ \frac{4}{3}\frac{\partial}{\partial T}\frac{u}{T} &= \frac{\partial}{\partial V}\frac{Vu'}{T} \end{aligned}$$

yielding

$$\frac{4}{3}\frac{u'T-u}{T^2}=\frac{u'}{T}$$
$$\frac{4}{3}u'T-\frac{4}{3}u=u'T$$
$$4u(T)=T\frac{du(T)}{dT}\Rightarrow u(T)=\sigma T^4$$

b) Entropy density $s=S/V$

$$S=\int 4\sigma VT^2dT=\frac{4}{3}\sigma VT^3$$
$$S=\int \frac{4}{3}\sigma T^3dV=\frac{4}{3}\sigma VT^3\checkmark$$

both are equivalent which is good.

$$s=\frac{4}{3}\sigma T^3$$

c) adiabatic & isochoric specific heat

Adiabatic: $S=\text{const.}$

$$C_V=\left(\frac{\partial U}{\partial T}\right)_V=4\sigma VT^3$$

d) Carnot cycle. Isothermal + Adiabatic cycle. Here are the relevant equations of the cycle that we'll need for part e.

$1\to 2$ Isothermal T_1	$Q=T_1\Delta S=-\frac{4}{3}\sigma T_1^4\left(V_1-V_2\right)$
$2\to 3$ Adiabatic	$V_3=V_2\left(\frac{T_1}{T_2}\right)^3$
$3\to 4$ Isothermal T_2	$Q=T_3\Delta S=-\frac{4}{3}\sigma T_2^4\left(V_2-V_3\right)$
$4\to 1$ Adiabatic	$V_1=V_4\left(\frac{T_2}{T_1}\right)^3$

Note that some work must be done during the adiabatic processes, but there is no heat transfer. The main difference with an ideal gas is the T^4 scaling instead of $U\propto T$, so there are no logarithms required.

e) show $\oint\frac{dQ}{T}=0$

Heat only flows during the Isothermal expansions, and we use the adiabatic steps to reduce the number of variables.

In [ ]:
```
from sympy import *
V = symbols('V_{0:5}', real=True, positive=True)
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T = symbols('T_{0:5}', real=True, positive=True)
boltzmann = symbols('sigma', real=True, positive=True)

Q_12 = -Rational(4,3) * boltzmann * T[1]**4 *(V[1]-V[2])
_23 = {V[3]: V[2]*(T[1]/T[2])**3}
Q_34 = -Rational(4,3) * boltzmann * T[2]**4 *(V[4]-V[3])
_41 = {V[1]: V[4]*(T[2]/T[1])**3}

remove_V3_V1 = _23 | _41

entropy1 = Q_12.subs(remove_V3_V1)/T[1]
entropy2 = Q_34.subs(remove_V3_V1)/T[2]
display(entropy1, entropy2)
Eq(entropy1, entropy2)
```

$$-\frac{4T_1^3\sigma\left(-V_2+\frac{T_2^3V_4}{T_1^3}\right)}{3}$$

$$-\frac{4T_2^3\sigma\left(-\frac{T_1^3V_2}{T_2^3}+V_4\right)}{3}$$

Out[ ]: True

5. Dieterici equation $p(v-b)=RTe^{-a/RT\nu}$

a) unstable for $\left(\frac{\partial p}{\partial \nu}\right)_T > 0$. I have no idea what this means physically. A tip from my extra special friend was to take the log of the equation.

$$\ln p + \ln(v-b) = \ln(RT) - \frac{a}{RT\nu}$$
$$\frac{\partial}{\partial \nu}\ln p + \frac{\partial}{\partial \nu}\ln(v-b) = -\frac{\partial}{\partial \nu}\frac{a}{RT\nu}$$
$$\frac{1}{p}\left(\frac{\partial p}{\partial \nu}\right)_T + \frac{1}{\nu-b} = \frac{a}{RT\nu^2}$$
$$\left(\frac{\partial p}{\partial \nu}\right)_T = \frac{ap}{RT\nu^2} - \frac{p}{\nu-b}$$

Which is positive for

$$\frac{ap}{RT\nu^2} > \frac{p}{\nu-b} > 0$$
$$\frac{a}{RT\nu^2} > \frac{1}{\nu-b} > 0$$

b) we need to boil the above down even more to find T_c

$$RT\nu^2 - a\nu + ab < 0$$

This is a quadratic equation, so the qadratic formula says real roots occur for

$$a^2 - 4RTab > 0$$
$$T_c < \frac{a}{4bR}$$

Using the same quadratic formula we have

$$\nu_c = \frac{-(-a)}{2RT_c} = \frac{a}{2R(a/4bR)} = 2b$$
$$p_c = R\left(\frac{a}{4bR}\right)e^{-2}$$
$$p_c = \frac{a}{4b^2}e^{-2}$$

c) $p \to pp_c$, $\nu \to vv_c$, $T \to TT_c$. I'll just chuck this one in wolfram.

$$p\frac{a}{4b^2}e^{-2}\left(2bv - b\right) = RT\frac{a}{4bR}e^{-\left(\frac{a}{RT(a/4bR)2bv}\right)}$$
$$p(2v-1) = Te^{2-\frac{2}{Tv}}$$

d) So, determine if the gas cools/warms while expanding?

$$\left(\frac{\partial T}{\partial p}\right)_H$$

with fixed enthalpy $H = U + pV$

$$dH = \left(\frac{\partial H}{\partial T}\right)_p dT + \left(\frac{\partial H}{\partial p}\right)_T dp$$
$$= C_p dT + \left(\frac{\partial H}{\partial p}\right)_T dp = 0$$
$$\to \left(\frac{\partial T}{\partial p}\right)_H = -\frac{1}{C_p}\left(\frac{\partial H}{\partial p}\right)_T$$

Then using $\left(\frac{\partial H}{\partial p}\right)_T = T\left(\frac{\partial S}{\partial p}\right)_T + v = -T\left(\frac{\partial v}{\partial T}\right)_p + v$ we have

$$\left(\frac{\partial T}{\partial p}\right)_H = \frac{1}{C_p}\left[T\left(\frac{\partial v}{\partial T}\right)_p - v\right]$$

Applied to Dietrici, we have

$$\left(\frac{\partial v}{\partial T}\right)_p = \frac{v(v-b)(RTv+a)}{T(RTv^2 - a(v-b))}$$

I'm sure the answer isn't far from here but I gotta study for this exam.

6. relations. Not sure what identies can be used to prove identites.

a) Use $\partial Q = T \partial S$

$$\frac{C_V}{T} = \left(\frac{\partial Q}{\partial T}\right)_V \Big/ T = \left(\frac{\partial S}{\partial T}\right)_V$$
$$\frac{C_p}{T} = \left(\frac{\partial Q}{\partial T}\right)_p \Big/ T = \left(\frac{\partial S}{\partial T}\right)_p$$

b) Maxwells relations. $dU = TdS - pdV$. These fall out of the double derivative thing.

$$\left(\frac{\partial}{\partial V}\left(\frac{\partial U}{\partial S}\right)_V\right)_S = \left(\frac{\partial}{\partial S}\left(\frac{\partial U}{\partial V}\right)_S\right)_V \quad \rightarrow \left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

The second one comes from Helmholtz $dF = dU - d(TS) = -SdT - pdV$